18

Feed Analysis

INTRODUCTION

The following section provides guidance on analytical methods that may be used to determine values needed for the equations and models in this publication. If alternate methods are used, particularly for the empirical assays (those assays such as neutral detergent fiber [NDF] that define the analyte measured), the user is advised to verify that the alternate gives results comparable to the cited method.

Reference is made to cutting or abrasion mills that may be used to process feed samples for analysis. Cutting mills (e.g., a Wiley mill) are those in which the samples are cut by the action of rotating and stationary steel knives. In an abrasion mill (e.g., Udy or Cyclone mills), samples are carried on air currents in a circular chamber and strike the abrasive interior of the chamber. A 2-mm screen in an abrasion mill gives a grind equivalent to a 1-mm screen in a cutting mill because of the angle at which the particle hits the screen.

Often the concentrations of assayed components in a feed do not sum to 100 percent. In the past, a fraction that was calculated by difference (e.g., nitrogen [N]-free extract or nonfiber carbohydrates) was included so the sum would equal 100 percent. However, when a by-difference fraction is not included, deviations from 100 percent can occur even with accurate execution of analytical methods and often are not a problem. One source of error is the conversion of N measure- ments to crude protein (CP). By default, N times 6.25 equals CP, although actual factors for plant and animal products range from 5.18 to 6.38 (Jones, 1931); nonprotein N sources such as urea can have much smaller factors. Some components can be found in multiple fractions and are counted more than once when summing. NDF can contain N and ash, and those por- tions will also be counted in the CP and ash fractions. There are also many analytes that are not detected by routine assays that will reduce recovery from 100 percent (e.g., pectins, tan- nins). Where possible, information on analytical error related to specific analyses is included but is not available for all methods.

Obtaining a representative sample of a feedstuff is crucial, and proper sampling procedures must be followed wherever subsampling is taking place, from the farm to the laboratory. The variability in composition data related to sampling can be large. Sampling variance for corn and haycrop silages accounted for 30 to 81 percent of within-farm variance with daily sampling and 9 to 37 percent with monthly sampling, with the variation varying by the analyte evaluated (St-Pierre and Weiss, 2015). A high level of sampling variation im- plies a low level of confidence that the analysis of a single sample reflects the composition of the feed. Taking multiple samples of silages over time and using rolling averages in ration formulation is recommended (Weiss et al., 2014). A challenge with feeds that are fed out rapidly but that may vary substantially in composition within or between lots is that analyses may only give a retrospective on feeds already consumed. A useful manual on factors that affect subsam- pling in the field and laboratory, as well as recommended techniques and tools, was published by the Association ol American Feed Control Officials (AAFCO, 2015). Practical recommendations for sampling silages and baled hay, with discussion on the issue of sampling variability, are found in Weiss et al. (2014).

CHEMICAL AND PHYSICAL ANALYSES

Dry Matter

Dry matter (DM) is usually determined by drying materi- als in a forced-air oven or microwave. Oven methods include 100 to 105°C overnight or to a constant weight, 135°C for 2 hours (AOAC Method 930.15). Drying samples at a lower temperature (55 to 60°C) in a forced-air oven for 24 hours before drying at 105°C for 48 hours can reduce the degree of browning and sample destruction that occurs. The portion of the feed that is lost in the drying process is referred to as "loss on drying."

Special Considerations

Samples intended for further analysis should not be dried at >60°C because of temperature-induced changes in protein solubility, lignin, and fiber concentrations (e.g., Van Soest, 1965). When large quantities of samples are dried, mixing the sample partway through the drying is important to fully dry the sample. Drying of samples at <60°C will result in lower DM values than those determined at higher temperatures. Use of a specific assay for water, such as a Karl Fischer titration, does not allow determination of DM by difference. The gen- erally used measures of DM or loss on drying remove water and other volatile substances and are different from specific measurement for water content.

Ash/Organic Matter

Ash and organic matter (OM) are commonly determined by combusting feeds in a muffle furnace. A variety of ap- proaches are used, including 600°C for 2 hours (AOAC Method 942.05), 500 to 600°C for 8 to 24 hours, and so on. The residue after ashing should be light gray to white in color. Other colors, including brown or black, are pos- sible because of the presence of minerals or formation of carbonates. If the residue is black, verify that the sample has been completely ashed by reashing, making sure that there is adequate oxygen admission to allow full combustion and possibly verifying the muffle furnace temperature.

Crude Protein/Nitrogen

N content of feeds is commonly analyzed by Kjeldahl analysis or Dumas combustion analysis.

Special Considerations

Assuming 16 percent of N in CP, 6.25 is the factor usually used to calculate CP from N. Although other factors may be more appropriate, depending on the N fractions and amino acid (AA) composition of the protein matrix (see Jones [1931] for additional factors), the 6.25 factor became almost universal, except for milk protein, for which 6.38 is commonly used. How- ever, based on the milk protein composition, a factor of 6.34 would be more appropriate (Karman and van Boekel, 1986). Measured components in blood meal often sum to > 100 percent likely because of an inappropriate N to CP conversion factor.

Soluble Protein

Soluble protein content of feeds is the difference between total CP and CP remaining after feed is extracted with a borate-phosphate buffer (Licitra et al., 1996). No differen- tiation is made between amino and nonprotein nitrogenous materials. Soluble protein is not used to determine nutrient supply in the present NRC.

Neutral Detergent-Insoluble Protein

Neutral detergent residue is prepared by processing a sam- ple as described for NDF analysis, with inclusion of sodium sulfite and heat-stable a-amylase in the extraction. The sample is then filtered under vacuum through filter paper, with subse- quent soaks and rinses with boiling distilled water and acetone as described in the NDF procedure. The residue and filter paper may be dried at 55 to 60°C prior to analysis for N by Kjeldahl or Dumas combustion analysis. The neutral detergent insoluble protein (NDIP) is expressed on a CP basis as N x 6.25.

Acid Detergent-Insoluble Protein

Acid detergent residue is prepared by processing a sample as described for acid detergent fiber (ADF) analysis. The sample is then filtered under vacuum through filter paper, with subsequent soaks and rinses as described in the ADF procedure. The residue and filter paper may be dried at 55 to 60°C prior to analysis for N by Kjeldahl or Dumas combus- tion analysis. The acid detergent insoluble protein (ADIP) is expressed on a CP basis as N x 6.25.

Amino Acids

Determination of the AA content of proteins requires their hydrolysis. Protocols may be found in the Methods of Analysis (AOAC, 2006, or later), the research literature, and other resources (e.g., Rutherfurd and Gilani, 2009). Briefly, concentrations of all AAs in proteins are determined after hydrolysis in hydrochloric acid (HC1), except tryptophan (Trp), which requires an alkaline hydrolysis, with the dura- tion of hydrolysis usually varying between 21 and 24 hours. Sulfur AAs, cysteine (Cys) and methionine (Met), must be protected prior to hydrolysis: the recommended procedure is a performic acid oxidation. Because the committee incorpo- rated AA requirements into the model, special attention has been devoted to obtaining the true AA composition of each protein fraction to be used in the model.

Correction of Amino Acid Composition of Protein

Two general points can be extended to all proteins when assessing the AA composition obtained from protein hy- drolysis. First, during hydrolysis, one molecule of water is added for each peptide bond that is cleaved. Therefore, 1 kg of pure protein perfectly hydrolyzed should yield ap- proximately 1.15 kg of free AAs, the ratio varying slightly depending of the AA composition of the protein. However, the hydrolysis of 1 kg of protein usually yields less than 1 kg of free AA, and that difference is often incorrectly assumed to equal nonamino (or nonprotein) N compounds. Indeed, some AAs are not stable in acid conditions, whereas other AAs are not completely released from the protein structure because peptide bonds involving hydrophobic residues are not all hydrolyzed by 24 hours. In practice, the 21- to 24-hour period has been chosen for practicality and laboratory cost to optimize the recovery of AAs that are labile in acid and of those AAs more resistant to hydrolysis (Robel and Crane, 1972; Rowan et al., 1992; Rutherfurd et al., 2008). Although this practice is usually adequate to rank feed ingre- dients based on AA composition, it limits the development of a factorial approach for AA supply and demand. On the supply side, the AA composition is estimated using results from hydrolyses with the abovementioned constraints. On the other hand, the requirement side includes proteins for nonproductive functions with a composition determined by hydrolysis and milk proteins, the major protein secretion, with an AA composition based on the residues obtained from DNA sequences of the different milk proteins.

In an attempt to improve the determination of AA com-position of protein, multiple time point hydrolyses had been performed on pure proteins (Robel and Crane, 1972; Darragh et al., 1996), diets and digesta (Rowan et al., 1992), and goat milk (Rutherfurd et al., 2008). The number of hydrolysis times varied from 4 to 19, and periods of hydrolysis varied from 2 to 168 hours. The best estimates of AA composition were either the maximal value of each AA obtained or esti- mated using an equation that included hydrolysis rate and the loss rate of the labile AAs once they were released into the HCI. Based on a single set of analyses, Rowan et al. (1992) proposed different corrections factors for each type of sample. However, based on a literature search and on recent work, there is no clear indication that different correction factors should be used for each type of matrix (Lapierre et al., 2019), although this is worthy of further investigation. Therefore, a single set of correction factors, one for each AA, is proposed and has been adopted in this edition (see Table 18-1) to correct the AA composition of all types of proteins reported following 24-hour hydrolysis. The AA compositions reported in the feed tables in this current publication are as analyzed and reported without application of the correction factors. The implications of this are discussed in Chapter 6. In the text, if the AA compo- sition is corrected for the incomplete recovery after a 24-hour hydrolysis, the concentration will be indicated as AA_{corr}. Note that once corrected with the proposed correction factors for incomplete recovery, the sum of the AA concentrations of a pure protein will yield approximately 1.15 times the weight of the hydrolyzed protein due to the addition of water. Some models (e.g., NorFor, 2011) correct the concentrations of each AA by the ratio of the molecular weight (MW) of the anhy- drous AA (MW of the AA—18, the MW of water) relative to the MW of the free AA. That way, 1 kg of pure protein yields 1 kg of anhydrous AA. Using such a scenario, one would have to back-calculate from anhydrous AA to "hydrated" AA for various derivations (e.g., to determine the amount of an AA to be supplied as a rumen-protected AA). Therefore, to avoid confusion, the fluxes of AA reported by the model will always refer to the free "hydrated" AA.

TABLE 18-1 Correction Factors Proposed for Individual AAs to Estimate the True AA Concentration from Concen- trations Obtained After a 24-Hour Hydrolysis[a]

Amino Acid	24-Hour Hydrolysis Correction Factor
Alanine	1.04
Arginine	1.06
Asparagine + aspartic acid	1.02
Cysteine	1.15
Glutamine + glutamic acid	1.05
Glycine	1.10
Histidine	1.07
Isoleucine	1.12
Leucine	1.06
Lysine	1.07
Methionine	1.05
Phenylalanine	1.06
Proline	1.04
Serine	1.12
Threonine	1.07
Tryptophan	1.06
Tyrosine	1.05
Valine	1.10

[a]Adapted from Lapierre et al., 2019.

Neutral Detergent Fiber

NDF is determined by gravimetric analysis of ground samples boiled under reflux for 1 hour in neutral detergent with heat-stable a-amylase and sodium sulfite, as well as filtered through coarse porosity crucibles (e.g., AOAC Method 2002.04; Mertens, 2002). Neutral detergent residues are dried and weighed, with NDF equal to the dry residue weight divided by the dry weight of the original sample. In this assay, a-amylase removes starch contamination and sodium sulfite solubilizes protein, thus reducing protein content of the residue. Selection of whether samples are expressed on a "with ash" or "ash-free" basis is dictated by the requirements of the application in which the values are used. Biogenic silica is solubilized by neutral detergent, but silica in soil is more resistant to solubilization and will contaminate the fiber residue. The analytical error (standard deviation) for replicate samples run in the same analytical run was approximately 0.4 percent for forages, 0.6 percent for concentrates < 10 percent fat, 0.4 percent for concentrates >10 percent fat, and 0.5 percent overall, on an as-received basis (Mertens, 2002).

Special Considerations

Ash present in NDF is erroneously counted as OM if NDF is not reported on an ash-free basis. This is typically a small error, unless samples are contaminated with soil. Soil-contaminated samples should be evaluated on an ash-free basis or the original materials resampled. In most applications of the current model, NDF is not on an ash-free basis because the base literature from which requirements were derived did not report NDF on an ash-free basis.

Samples with >10 percent fat should be preextracted with acetone (Mertens, 2002) or NDF values may be inflated as the detergent solubilizes in a lipid layer during extraction.

The original method calls for grinding samples through the I -mm screen of a cutting mill (e.g., a Wiley mill; Goering and Van Soest, 1970). Using a finer grind size (e.g., abrasion mill with 1-mm screen) may reduce NDF values if finer material passes through the filter used. The AOAC Method 2002.04 recommends use of sand or glass fiber filters as filtration aids. This can help greatly with difficult-to-filter samples.

Acid Detergent Fiber

ADF is determined by gravimetric analysis of ground samples boiled under reflux for 1 hour in acid detergent and filtered through coarse porosity crucibles (AOAC Method 973.18). Acid detergent residues are dried and weighed, with ADF equal to the dry residue weight divided by the dry weight of the original sample.

Special Considerations

Ash in the ADF residue is erroneously counted as OM if ADF is not reported on an ash-free basis, but as with NDF, the error is usually small unless samples are contaminated with soil or contain substantial amounts of biogenic silica. Soil-contaminated samples should be evaluated on an ash-free basis. Biogenic silica, the silica naturally incorporated into the structure of plants, is insoluble in acid detergent and will con- taminate the acid detergent residue. Biogenic silica is found in both C3 and C4 grasses, including sugar cane. Forages suspected to contain higher levels of silica should be evalu- ated on an ash-free basis to avoid treating ash as carbohydrate.

Samples with >10 percent fat should be preextracted with acetone (Mertens, 2002) or ADF values may be inflated as the detergent solubilizes in a lipid layer during extraction.

When analyzing for ADF, care should be taken to boil samples for no more than 1 hour and to fully soak and wash all residual reagent from the residue. The acid nature of the reagent will remove more of the sample with longer boiling or further degrade the sample during drying. The original method calls for grinding samples through the 1-mm screen of a cutting mill (e.g., a Wiley mill; Goering and Van Soest, 1970), and finer grinding may reduce ADF concentrations.

Lignin

Lignin is determined as sulfuric acid lignin analyzed on acid detergent fiber residues (AOAC Method 973.18; AOAC, 2006) .

Special Considerations

The 72 percent sulfuric acid used in this assay should be chilled to 15°C before use and crucibles maintained at 20 to 23°C. Failure to do so will result in increased solubilization of the sample and reduced measured lignin values.

Starch

Starch is commonly analyzed using enzymatic-colorimetric methods (e.g., Bach Knudsen, 1997, AOAC Method 2014.10; Hall, 2015). In these methods, the starch is gelatinized using heat and moisture, or alkali, followed by hydrolysis with enzymes specific to the α-1,4 and α-1,6 linkages in starch. A combination of heat-stable α-amylase (E.C. 3.2.1.1) and amyloglucosidase (glucoamylase; E.C. 3.2.1.3) or amylo- glucosidase alone is used. If heat and moisture are used for gelatinization, use of acidified buffers is recommended to avoid the conversion of a portion of the starch to maltulose (Dias and Panchal, 1987), which reduces starch recovery. Specific measurement of glucose with methods such as the glucose oxidase-peroxidase assay (e.g., Karkalas, 1985; McCleary et al., 1997), high-performance liquid chromatog- raphy (HPLC), or gas chromatography (GC) is recommended to avoid interference from other carbohydrates. The total glucose measured in enzymatically treated samples minus the free glucose in the original sample equals glucose from starch, and that glucose x 0.9 equals the amount of starch. The 0.9 factor reflects the ratio (162/180) of anhydroglucose (162 g/mol) to glucose (180 g/mol) after removal of one water molecule (18 g/mol) added during hydrolysis per glucose molecule bound into the polysaccharide chain. Glucose car- ried through the assay should give a value of 88 to 92 percent starch; pure sucrose should give a value less than 1 percent. The analytical error (standard deviation) for replicate samples of livestock feeds in the same analytical run was approxi- mately 0.1 percent for low-starch feeds and 0.3 percent for high-starch feeds on an as-received basis (Hall, 2015).

Special Considerations

Samples should be ground to pass the 1 -mm screen of an abrasion mill, the 0.5-mm screen of a cutting mill, or a 40-mesh screen. Use of alkali in the analysis may include starch that is resistant to digestion by mammalian enzymes in the total starch value, depending on the run conditions (McCleary et al., 2002). Use of enzyme preparations that release glucose from nonstarch carbohydrates inflates starch values. If analy- sis of sucrose and cellulose samples yields values greater than 1 percent glucose, the enzymes or assay conditions are releasing glucose from these nonstarch substrates. Do not use recovery values of starch control samples to adjust the starch values of other samples. This approach assumes that all samples behave similarly to the control samples, which may be incorrect.

In feeds including but not limited to cooked starchy feeds, bakery or candy by-products, and some corn hybrids, a portion of the starch may be hydrolyzed to material that is water soluble but analyzes as starch. Accordingly, it may be counted in both the starch fraction and water-soluble car- bohydrates (WSC). A potential solution to this issue is to (1) use a starch assay to measure the starch in the water extract used to determine WSC and subtract it from the starch value or (2) determine the recovery factor for solubilized starch and maltooligosaccharides in the phenol-sulfuric acid assay to allow correction of the WSC for the solubilized starch. The recovery factor is determined by analyzing several different concentrations of maltooligosaccharides or solubi- lized starch using the usual standard carbohydrate (usually sucrose) for the WSC assay and determine the recovery of starch or maltooligosaccharides as recovery = detected g/ actual g. Multiply the measured starch in the water extract by 1/recovery to estimate the portion of the WSC attributable to water-soluble starch. Correct the total starch or WSC for the solubilized starch. Determinations of total starch digestibility will include the water-soluble starch.

Water-Soluble Carbohydrates

WSC are determined by extraction of samples with water and analysis of the extract for carbohydrates (e.g., Faithfull, 2002; Uden, 2006; Hall, 2015). The water will solubilize monosaccharides (glucose, fructose, etc.), disaccharides (sucrose, lactose, maltose), oligosaccharides (stachyose, raffinose, maltooligosaccharides), and fructans. Use of a broad-spectrum carbohydrate detection method such as the phenol-sulfuric acid assay (Dubois et al., 1956) is prefer- able to a reducing sugar assay, particularly when samples high in fructans or lactose are analyzed (Hall, 2013,2014) as fructans will be overestimated and lactose underestimated. The carbohydrate chosen as the standard for the detection method affects the WSC values. Different carbohydrates give different responses in both the phenol-sulfuric acid assay (Dubois et ah, 1956) and reducing sugar assays (Weinbach and Calvin, 1935). Sucrose is commonly used as a standard in the detection method as it reflects the primary carbohydrate present in WSC in many feeds. However, if the analyst knows that a carbohydrate other than sucrose predominates, such as lactose in milk products or fructans in cool-season grasses, the carbohydrate that predominates should be used as the standard to obtain more accurate values.

Special Considerations

Samples should be ground to pass the 1-mm screen of an abrasion mill, the 0.5-mm screen of a cutting mill, or a 40-mesh screen. The phenol-sulfuric acid assay (Dubois et ah, 1956), which uses 0.5 mL each of sample solution and 5 percent phenol solutions and 2.5 mL of acid, gives more consistent results in workable volumes than the 80 percent phenol solution used by Dubois et ah (1956). The sample solution should be analyzed in duplicate with vortexing after each liquid addition.

Bakery products may give higher WSC values than ex- pected because some of the starch hydrolyzes to maltooligo- saccharides during baking. Maltooligosaccharides or soluble starch are water soluble and will analyze with the WSC. Analysis of the water extract using a starch assay to detect enzyme-released glucose minus free glucose will show what portion of the WSC is solubilized starch and maltooligosac- charides. See section on starch for details.

The values for ethanol-soluble carbohydrates (80 percent ethanol extraction) are similar to water-soluble carbohydrate values except for feeds containing substantial amounts of carbohydrate that are extracted by water but not ethanol, such as cool-season grasses with fructans and products containing lactose. Molasses products are typically analyzed for “total sugars as invert,” and this value is given on the feed tag. Given that other WSC such as lactose have not been blended with the molasses, total sugars as invert may be used for the WSC value in molasses products.

Residual Organic Matter

Residual organic matter (ROM) is the fraction not ac- counted for by the major nutrients and is calculated (see Equation 3-1) by subtracting crude protein (CP), starch, NDF, and fatty acids (FAs) from OM. It contains water- soluble carbohydrates, ingested fermentation, and other short-chain FAs (such as acetic, lactic butyric acids, plant organic acids), glycerol (both free and the glycerol moiety of triglycerides), soluble fiber (e.g., pectins, gums), any other components not accounted for in the main feed fractions (e.g., tannins, waxes, pigments), and analytical error associ- ated with determination of the main feed fractions. It is used for the calculation of energy values.

Fats/Fatty Acids

FA content of feeds is determined by GC of sample extracts in which all FAs present have been converted to methyl esters (Sukhija and Palmquist, 1988). Simultaneous methylation and extraction in this procedure more com- pletely extracts FAs from feeds than does preextraction and subsequent methylation. The diverse lipid profiles in samples may recommend method variants—benzene, chloroform, or elevated temperature—to achieve more complete extraction and precision of analyses (Sukhija and Palmquist, 1988). Use of crude fat is not recommended as a measure of nutrition- ally useful lipid. In addition to FAs, crude fat (CF) contains plant waxes, cutin, pigments, and indigestible materials of similar solubilities. The use of CF as a percentage of DM minus 1 to estimate FA content in feeds (NRC, 2001) is no longer recommended because it can overestimate FAs by up to 2 percent of DM in forages and mixed diets (Palmquist and Jenkins, 2003). Because of feed labeling, some feeds must be analyzed for CF and should be done via the AOAC Official Method 2003.05 (AOAC, 2006).

Special Considerations

Quantitative extraction of lipids prior to FA analysis is critical. Internal standards of FAs that do not occur in feeds should be added to adjust for loss of FAs during the analytical process (Palmquist and Jenkins, 2003; Jenkins, 2010). When extraction of lipids is followed by FA analysis, there is no concern about using solvent systems that also extract non-FA components, provided that these compounds do not subsequently manifest as unidentified FAs in the chromatogram and erroneously contribute to total FAs (Alves and Bessa, 2007).

Mid-infrared (MIR) analysis has been used to analyze FAs in milk (Soyeurt et al., 2011), and it is more accurate for major than for minor FAs, and accuracy was lower for MIR predictions of polyunsaturated FAs, n-3 FAs, and branched-chain FAs. The MIR predictions of FA content in milk (g/L milk) may be more accurate than those developed for FA determinations in milk fat (Soyeurt et ah, 2011). See Jenkins (2010) for additional insights on factors that can affect FA analysis.

Minerals

Mineral analyses are commonly carried out with induc- tively coupled plasma mass spectrometry or atomic absorp- tion, although there are also chemical gravimetric or colo- rimetric assays available for some minerals. Mineral values reported from near-infrared reflectance spectroscopy (NIRS) analyses are based on correlations of minerals with certain organic components of feeds. Because NIRS does not directly measure minerals, the results cannot be considered definitive.

Special Considerations

Chromium (Cr)—Samples processed through harvesting equipment and grinders with steel or chromed components will be contaminated with Cr from those sources; soil contamination may also elevate sample Cr (Spears et ah, 2017). It appears that the degree of contamination is variable. Samples should be ground with a ceramic device.

Phosphorus (P)—For the current model, feed P is analytically partitioned into organic P and inorganic P. Organic P is calculated as total P - inorganic P. Total P can be determined spectrophotometrically by a molybdovanadate method (AOAC Method 965.17; AOAC, 2006) in addition to other spectroscopic methods. Inorganic P can be determined by extraction of samples with 0.5 M HC1 and detection of P in the extract (Ray et ah, 2012).

Particle Size

Use of the Penn State Particle Separator is recommended for use with the physically adjusted NDF (paNDF) system (Kononoff et ah, 2003; Heinrichs, 2013). Four separator boxes with 19 mm (0.75 in.), 8 mm (0.31 in.), 4 mm (0.16 in.), and the solid pan are used, stacked from the largest sieve opening through the smallest with the pan on the bottom. Approximately 1.4 L of total mixed ration or silage is placed on the top screen. Boxes are shaken vigorously on a flat sur- face five times in one direction, then turned one-quarter turn, and shaken again. The 5 shaking reps are repeated a total of eight times to give a total of 40 shakes. The shakes should be done at a rate of 1.1 shakes per second over a distance of 17 cm (7 in.). Once completed, the material on each sieve is weighed. The DM on each sieve is determined using microwave, forced-air oven, or other suitable method. The percentage of the total sample DM on each sieve is determined.

DIGESTION ANALYSES

In Vitro Neutral Detergent Fiber Digestibility

Samples are anaerobically incubated in vitro with ruminal inoculum per Goering and Van Soest (1970). Samples for NDF are ground to pass a 1-mm screen of a cutting mill and are incubated for 30 or 48 hours for NDF. For neutral detergent fiber digestibility (IVNDFD), the entire contents of the flask are analyzed according to the method for NDF and the residue recovered through filtration under vacuum. Digestibility of NDF at a given time point is calculated on a DM basis as follows: (NDF as percentage of original feed - residual NDF as percentage of original feed) / NDF as percentage of original feed.

In Situ Measurement of A, B, and C Protein Fractions

Samples are ground to pass the 2-mm screen of a cutting mill. Samples are weighed into 40- to 60-|im pore size dacron/polyester bags that are then sealed. The ratio of sample weight to surface area of the bag should be 10 mg/cm² (Vanzant et al., 1998). Bags are presoaked in water and then incubated in the ventral rumens of at least two ruminally cannulated animals fed characterized diets, ideally including the feed- stuff being incubated. Incubations are for at least five time points of incubation that include at least 0 and 48 hours (and another time of 72 hours for forages) such that a nonlinear mathematical model can fit the washout and extent fractions and the resultant k_d of the B pool. Endpoints <48 hours are problematic because for many rumen-undegradable protein (RUP) sources, the C fraction could not be resolved (Liebe et al., 2018). The committee recommends that a standard substrate be included. After removal, samples are subject to a consistent and thorough rinsing in a washing machine for five 1-minute rinses in cold water. Correction for microbial contamination is highly recommended, particularly for for- ages (Krawielitzki et al., 2006; Kamoun et al., 2014) and fibrous by-products (Paz et al., 2014). Failure to correct for microbial contamination in incubated residues can reduce es- timates of ruminally degradable protein (Alexandrov, 1998). Multiple models have been evaluated for in situ disappear- ance (Lopez et al., 1999), but the committee recommends a first-order model for consistency of protein degradation and to use blank bags to assess infiltration of particulate matter.

Special Considerations

Loss of nonsoluble matter in the form of small particles from the in situ bags can inflate the value of the A fraction and reduce that of the B or C fraction (Maxin et al., 2013). Methods have been applied to address this issue (Maxin et al., 2013), but more work is needed to assess the ap- proaches and the applicability of B fraction rates of digestion and C fraction indigestibility to the small particles.

Standard substrates have been included in in situ incu- bations to allow covariate adjustment of the results across incubation runs. However, this approach assumes that the other substrates in the incubation behave like the standard, that there is a body of data from multiple runs using the same standard for comparison, and that A, B, and C fractions can be adjusted to give accurately comparable values among runs. These assumptions may not be correct. The use of stan- dards to flag incubations that have abnormally high or low digestibilities relative to other runs can be used to determine whether results from a total run should be discarded. The stan- dard must be subsampled accurately and stored appropriately to avoid changes in composition and digestibility over time.

As stated previously (NRC, 2001), standardized approaches are strongly recommended, but often they are not stringently followed (Liebe et al., 2018). Hence, certain inherent assump- tions or limitations should be reiterated. Decreasing particle size of ground forages reduces subsampling error, but the fractional k_d increases with decreasing particle size (Michalet- Doreau and Cemeau, 1991). Grinding through a 2-mm sieve appears to improve homogeneity of samples while minimizing particle losses of NDF (Krizsan et al., 2015) and presumably N. In the committee's database, forage processing ranged from being cut by scissors (especially for fresh forage simulating that grazed) to being ground through a 1 - or 2-mm screen while frozen or after oven drying at <60°C. Bacterial contamination in residues from ruminal incubation is well known (NRC, 2001; Krawielitzki et al., 2006). This issue led to standardization for forages (Klopfenstein et al., 2001), but similar problems ex- ist and deserve correction for fibrous by-products (Paz et al., 2014). When NDF exceeds 20 percent in a feedstuff, the com- mittee recommends that studies correct for bacterial contami- nation. Infusion of ^{15}N followed by recovery of ^{15}N enrichment in particulate-phase bacteria and residual feed in the bags is one potential approach (Kamoun et al., 2014), but that is costly and ^{15}N enrichment might be stratified in the rumen. Assaying the residue and bacteria using quantitative polymerase chain reac- tion (qPCR) offers promise (Paz et al., 2014).

Intestinal Digestibility of Rumen-Undegradable Protein

Both in vitro and mobile bag techniques are used for determining intestinal digestibility of RUP of feedstuffs. Pre- incubation with ruminal inoculum is needed for some feeds prior to mobile bags being introduced into the duodenum (Hvelplund, 1985) or prior to the intestinal digestibility step in the original three-step procedure (Calsamiglia and Stern, 1995). Many laboratories have adopted the modified three- step procedure (Gargallo et al., 2006) for in vitro assessment, which has been modified to use small pore-size filter paper to recover undegraded but soluble proteins (Hristov et al., 2019). Although early intestinal digestibility values were derived from mobile bags retrieved from an ileal cannula to reduce error associated with bacterial contamination as bags passed through the colon, virtually all of the subsequent values since NRC (2001) were obtained with bags retrieved from the feces. An alternative in vitro method for measuring small intestinal digestibility of AA in RUP is that of Ross (2013). As of this writing, peer-reviewed publications on the method or com- parison of its results to in vivo small intestinal digestibilities were not available. The results of this method were not used in the development of the nutritional model in this publication and have not been evaluated or tested for generating inputs for the National Academies of Sciences, Engineering, and Medicine model. Accordingly, the committee cannot provide recommendations for their use at this time.

Special Considerations

Procedures for measuring digestibility of RUP protein (dRUP) need to be standardized and further assessed for interactions with feed type. For example, pore size of the bag affects the digestibility values and likely pore size and substrate interact (Jarosz et al., 1994; Liebe et al., 2018). Insufficient comparisons have been made with in vivo ap- proaches, particularly because of the limitation of using ileal cannulas in dairy cows (Hristov et al., 2019). As with ruminal incubations, dRUP assays also should have correction for bacterial N contamination, particularly if the feedstuff has residual NDF (Jarosz et al., 1994).

Bioavailability of Rumen-Protected Amino Acids

Rumen-protected AA (RP-AA) supplementation decisions require knowledge of the bioavailability of the AA from all ingredients in the diet, including the RP-AA. Commentary on these methods is provided in Chapter 6. There is no clear con- sensus about which method should be used; different methods can give different values. Different methods to estimate the bioavailability of commercial RP-AA have been extensively detailed in Whitehouse (2016). These methods can be cat- egorized as in vitro, in situ, or in vivo methods (Whitehouse et al., 2017). In vitro methodology relies on incubation of the RP-AA in rumen fluid collected from cows or in a buf- fer mimicking the rumen environment followed by in vitro exposure to conditions simulating the intestinal digestion (Calsamiglia and Stem, 1995). The in vitro method obviously lacks the effect of animal and does not include any interaction of the RP-AA with the other feed ingredients of the diet. The in situ methods rely on estimation of rumen and intestinal dis- appearance of RP-AA in polyester or dacron bags. Ruminal incubation is for a fixed time period thought to reflect mean exposure when fed. Bags containing the ruminal incubation residue are subsequently introduced into a duodenal cannula and recovered either at the ileum or in the feces (e.g., Jarosz et al., 1994; Overton et al., 1996; Berthiaume et al., 2000). The in situ method also does not expose the RP-AA to the feed ingredients or handling, lacks the effect of chewing and ruminating, and assumes that all particles leaving the bag are degraded. Sometimes, these methodologies are mixed—for example, an in situ rumen incubation followed by an in vitro estimation of intestinal digestibility (e.g., Bach and Stem,

2000) . These two methodologies do not consider factors af- fecting the residence time of the RP-AA in the rumen, which affects degradation of certain types of RP-AA. For example, rumen degradation ranged from 9 to 37 percent for residence times varying from about 2 to 12 hours for an ethyl-cellulose protected product (Berthiaume et al., 2000).

In vivo methodologies are based on blood concentrations, labeled AA dilution, or production responses when the RP-AA is fed or introduced into the rumen. For blood concentration and production responses, the results need to be compared to either a base diet or the response to a known quantity of absorbed AA. Ideally, the known quantity is a postruminal infusion of the AA (e.g., Graulet et al., 2005; Koenig and Rode, 2001), but in some studies, the comparison was to an RP-AA with known bioavailability. Plasma concentration of lysine (Lys) and Met increased linearly with increasing levels of postrumen infusion of each AA (Whitehouse, 2016) up to at least 85 g/d of Lys (Borucki Castro et al., 2008) or 30 g/d of DL-Met (Rulquin and Kowalczyk, 2003). However, King et al. (1991) reported that at 180 g/d Lys, the relationship was strongly quadratic, whereas Lapierre et al. (2012) reported that at 15 g/d of DL-Met infused postruminally, the relation- ship was quadratic but with a significant linear component. Recommended methodologies are available (Whitehouse, 2016; Whitehouse et al., 2017). Key points include the use of both dietary and postrumen infusion of the test AA at rates similar to expected supplementation. The basal diet should be just adequate in MP and the AA of interest. To improve the accuracy of the method, Whitehouse (2016) proposed to express the concentration of the Lys relative to the total AA (excluding Lys) or the concentration of Met+Cys relative to the total AA (excluding the sulfur AA), rather than in absolute values of concentration, for RP-Lys and RP-Met, respectively.

The blood sampling schedule should at least cover one feeding interval, and sampling on more than one day should decrease the variability. The main advantage of this in vivo methodology is that the response obtained may include all of the factors that affect the bioavailability of the RP-AA. For example, me- chanical mixing of RP-Lys as well as exposure to a total mixed ration (the lower the DM, the higher the effect) increased the rumen in situ release of Lys; these effects were different depending on the type of RP-Lys (Ji et al., 2016). The main disadvantage is that the trials are expensive, and the resulting bioavailability can have high variance (20 percent or greater) for single-point assessments (Rulquin and Kowalczyk, 2003; Borucki Castro et al., 2008; Whitehouse, 2016). Whether this technique will work for multiple AAs delivered by a protein source is not known. In addition, this methodology does not determine whether the loss of the RP-AA occurs before the ingestion of the RP-AA, in the rumen or across the intestine.

Another response that has been used is milk protein yield (MPY), which is based on regression just as for the blood concentration method above and is subject to the same considerations regarding linearity and range. Because MPY response is an indirect measurement, the confidence interval may be large, and that could compromise the accuracy of the estimation of the bioavailability (Whitehouse, 2016). In ad- dition, one needs to ensure that the animals are deficient in the AA so that MPY response can be observed. This has been problematic in the past as the models have lacked precision, but the updated equations provided in this edition will hope- fully reduce this problem. This method also might be useful when evaluating Met analogues, which might not increase blood Met concentrations but might affect MPY.

Another method uses the blood concentration response when a large dose of the residue from a ruminally incubated RP-AA is infused into the abomasum to estimate intestinal availability. The area under the blood concentration curve resulting from the infused RP-AA is compared to an ab- omasally dosed, unprotected AA to derive a bioavailability value. Ruminal incubation is generally for a set time as for the in situ methods. This method has been shown to produce results equivalent to the regression method discussed previ- ously (Graulet et al., 2005). However, it likely becomes less accurate and more biased as the rate of release from the RP- form decreases. Adequate samples are needed to accurately calculate area under the curve and to better detect the start and end of the elevated concentrations. A second concern with this method is that the large dose of AA could stimulate catabolism. If the unprotected and protected doses are similar and absorption occurs over a similar time frame, the concern is likely not valid. However, the bioavailability and the rate of release of a naive RP-AA are unknown so that the dose of the unprotected AA may not be appropriate, which may necessitate a second trial. The method also relies on in situ incubations, which may not replicate the true ruminal effects. Graulet et al. (2005) observed consistent responses among the blood concentration regression methods and a pulse-dose response; however, bioavailabilities determined from milk protein responses were not consistent with those from the pulse-dose method (Fleming et al., 2019). This suggests that the method may rank RP-forms correctly, but the results may be biased in terms of the absolute bioavailability estimates.

Isotopic dilution methods have been used, and they pro- vide greater precision than other methods and are applicable to all feed ingredients (Borucki Castro et al., 2008; Estes et al., 2018; Huang et al., 2019). This approach relies on the plasma dilution of AA labeled with stable isotope (either ^{15}N or ^{13}C) administered into the bloodstream either continuously (Borucki Castro et al., 2008; Estes et al., 2018; Huang et al., 2019) or as a single dose (Borucki Castro et al., 2008; Maxin et al., 2013). Availability is derived by difference from a base diet or in comparison to provision of a known quantity of the AA. An extension of this method uses seleno-Met as a label with measurements of selenium concentrations in milk protein to deduce the dilution of seleno-Met and thus Met absorption (Weiss and St-Pierre, 2009). Results from these methods are promising as they are less invasive, are generally less expensive to conduct, are more precise (10 to 12 percent errors of determination for the ^{13}C method), and, for the ^{13}C- method, are broadly applicable across the essential AAs. The drawback is that they do not distinguish between ruminal and intestinal losses.

REFERENCES

AAFCO (Association of American Feed Control Officials). 2015. GOOD Samples: Guidance on Obtaining Defensible Samples. Champaign, IL: Association of American Feed Control Officials. https://www.aafco.org/Portals/O/SiteContent/Publications/GOODSamples.pdf. Accessed on June 12, 2019.

Alexandrov, A. N. 1998. Effect of ruminal exposure and subsequent microbial contamination on dry matter and protein degradability of various feedstuffs. Anim. Feed Sci. Technol. 71:99-107.

Alves, S. P., and R. J. B. Bessa. 2007. Identification of cw-12,ch-15 octadecadienoic acid and other minor polyenoic fatty acids in ruminant fat. Eur. J. Lipid Sci. Technol. 109:879-883.

AOAC (Association of Official Analytical Chemists). 2006. Official Methods of Analysis. 18th ed. Gaithersburg, MD: Association of Official Analytical Chemists.

Bach, A., and M. D. Stem. 2000. Measuring resistance to ruminal degradation and bioavailabiity of ruminally protected methionine. Anim. Feed Sci. Technol. 84:23-32.

Bach Knudsen, K. E. 1997. Carbohydrate and lignin contents of plant materials used in animal feeding. Anim. Feed Sci. Technol. 67:319-338.

Berthiaume, R., H. Lapierre, M. Stevenson, N. Coté, and B. W. McBride. 2000. Comparison of the in situ and in vivo intestinal disappearance of ruminally protected methionine. J. Dairy Sci. 83:2049-2056.

Borucki Castro, S. I., H. Lapierre, L. E. Phillip, P. W. Jardon, and R. Berthiaume. 2008. Towards non-invasive methods to determine the effect of treatment of soya-bean meal on lysine availability in dairy cows. Animal 2(2):224—234.

Calsamiglia, S., and M. D. Stem. 1995. A three-step in vitro procedure for estimating intestinal digestibility of protein in ruminants. J. Anim. Sci. 73:1459-1465.

Darragh, A. J., D. J. Garrick, P. J. Moughan, and W. H. Hendricks. 1996. Correction for amino acid loss during acid hydrolysis of a purified protein. Anal. Biochem. 236:199-207.

Dias, F. F., and D. C. Panchal. 1987. Maltulose formation during sacchari- fication of starch. Starch/Stdrke 39:64-66.

Dubois, M., K. A. Gilles, J. K. Hamilton, P. A. Rebers, and F. Smith. 1956. Colorimetric method for determination of sugars and related substances. Anal. Chem. 28:350-356.

Estes, K. A., R. R. White, P. S. Yoder, T. Pilonero, H. Schramm, H. Lapierre. and M. D. Hanigan. 2018. An in vivo stable isotope-based approach for assessment of absorbed amino acids from individual feed ingredients within complete diets. J. Dairy Sci. 101(8):7040-7060.

Faithfull, N. T. 2002. The analysis of animal feed and plant materials. Pp. 124-153 in Methods in Agricultural Chemical Analysis: A Practical Handbook. Wallingford, UK: CABI Publishing.

Fleming, A. J., K. A. Estes, H. Choi, B. A. Barton, C. A. Zimmerman, and M. D. Hanigan. 2019. Assessing bioavailability of ruminally protected methionine and lysine prototypes. J. Dairy Sci. 102(5):4014-4024.

Gargallo, S., S. Calsamiglia, and A. Ferret. 2006. Technical note: A modified three-step in vitro procedure to determine intestinal digestion of proteins. J. Anim. Sci. 84(8):2163—2167.

Goering, H. K., and P. J. Van Soest. 1970. Forage Fiber Analysis (Apparatus, Reagents, Procedures and Some Applications). Agriculture Handbook No. 379. Washington, DC: Agricultural Research Service, U.S. Depart ment of Agriculture.

Graulet, B., C. Richard, and J. C. Robert. 2005. Methionine availability in plasma of dairy cows supplemented with methionine hydroxy analog isopropyl ester. J. Dairy Sci. 88:3640-3649.

Hall, M. B. 2013. Efficacy of reducing sugar and phenol-sulfuric acid assays for analysis of soluble carbohydrates in feedstuffs. Anim. Feed Sci. Technol. 185:94-100.

Hall, M. B. 2014. Selection of an empirical detection method for determination of water-soluble carbohydrates in feedstuffs for application in ruminant nutrition. Anim. Feed Sci. Technol. 198:28-37.

Hall, M. B. 2015. Determination of dietary starch in animal feeds and pet food by an enzymatic-colorimetric method. Collaborative study. J. AOAC Int. 98:397-409.

Heinrichs, J. 2013. The Penn State particle separator. Pp. 1-8 in Penn State Cooperative Extension Bulletin. DSE 2013-186.

Hristov, A. N.. A. Bannink, L. A. Crompton, P. Huhtanen, M. Kreuzer, M. McGee, P. Nozierc, C. K. Reynolds, A. R. Bayat, D. R. Yanez-Ruiz, J. Dijkstra, E. Kebreab, A. Schwarm, K. J. Shingfield, and Z. Yu. 2019. Invited review: Nitrogen in ruminant nutrition: A review of measurement techniques. J. Dairy Sci. 102:5811-5852.

Huang, X., K. A. Estes, P. S. Yoder, C. Wang, N. Jiang, T. Pilonero, and M. D. Hanigan. 2019 Assessing availability of amino acids from various feedstuffs in dairy cattle using a stable isotope-based approach. J. Dairy Sci. 102:10983-10996.

Hvelplund, T. 1985. Digestibility of rumen microbial protein and undegraded dietary protein estimated in the small intestine of sheep and by in sacco procedure. Acta Agric. Scand. 25:132-144.

Jarosz, L., T. Hvelplund, M. R. Weisbjerg, and B. B. Jensen. 1994. True digestibility of protein in the small intestine and the hind gut of cows measured with the mobile bag technique using 15N-labelled roughage. Acta Agric. Scand. 44:146-151.

Jenkins, T. C. 2010. Technical note: Common analytical errors yielding inaccurate results during analysis of fatty acids in feed and digesta samples. J. Dairy Sci. 93:1170-1174.

Ji, P., H. A. Tucker, R. E. Clark, M. Miura, and C. S. Ballard. 2016. Short communication: Effect of on-farm feeding practices on rumen protected lysine products. J. Dairy Sci. 99(2): 1242-1246.

Jones, D. B. 1931. Factors for Converting Percentages of Nitrogen in Foods and Feeds into Percentages of Proteins. Circular No. 183. Washington, DC: U.S. Department of Agriculture

Kamoun, M,, H. Ammar, A. Thdwis, Y. Beckers, J. France, and S. Lopez. 2014. Comparison of three 15n methods to correct for microbial contamination when assessing in situ protein degradability of fresh forages. J. Anim. Sci. 92:5053-5062.

Karkalas, J. 1985. An improved enzymic method for the determination of native and modified starch. J. Sci. Food Agric. 36:1019-1027.

Karman, A. H,, and A. J. S. van Boekel. 1986. Evaluation of the Kjeldahl factor for conversion of the nitrogen content of milk and milk products to protein content. Netherlands Milk Dairy J. 40:315-336.

King, K. J., W. G. Bergen, C. J. Sniffen, A. L. Grant, D. B. Grieve, V. L. King, and N. K. Ames. 1991. An assessment of absorbable lysine re- quirements in lactating cows. J. Dairy Sci. V4(8):2530—2539.

Klopfenstein, T. J., R. A. Mass, K. W. Creighton, and H. H. Patterson. 2001. Estimating forage protein degradation in the rumen. J. Anim. Sci. 79(E. Suppl.):E208-E217.

Koenig, K. M., and L. M. Rode. 2001. Ruminal degradability, intestinal disappearance, and plasma methionine response of rumen-protected methionine in dairy cows. J. Dairy Sci. 84:1480-1487.

Kononoff, P. J., A. J. Heinrichs, and D. R. Buckmaster. 2003. Modifica- tion of the Penn State forage and total mixed ration particle separator and the effects of moisture content on its measurements. J. Dairy Sci. 86:1858-1863.

Krawielitzki, K., T. Schmidt, J. Voigt, J. Kowalczyk, and M. Gabel. 2006. Dynamics of microbial contamination of protein during ruminal in situ incubation of feedstuffs. J. Anim. Feed Sci. 15:313-328.

Krizsan, S. J., M. Rinne, L. Nyholm, and P. Huhtanen. 2015. New recom- mendations for the ruminal in situ determination of indigestible neutral detergent fibre. Anim. Feed Sci. Technol. 205:31-41.

Lapierre, H., G. Holtrop, G. Calder, J. Renaud, and G. E. Lobley. 2012. Is D-methionine bio-available to the dairy cow? J. Dairy Sci. 95:353-362.

Lapierre, H., S. Binggeli, M. Sok, D. Pellerin, and D. R. Ouellet. 2019. Esti- mation of correction factors to determine the true amino acid concentra- tion of protein after a 24-hour hydrolysis. J. Dairy Sci. 102:1205-1212.

Licitra, G., T. M. Hernandez, and P. J. Van Soest. 1996. Standardization of procedures for nitrogen fractionation of ruminant feeds. Anim. Feed Sci. Technol. 57:347-358.

Liebe, D. M,, J. L. Firkins, H. Tran, P. 3. Kononoff, and R. R. White. 2018. Technical note: Methodological and feed factors affecting measure- ment of protein A, B. and C fractions, degradation rate, and intestinal digestibility of rumen-undegraded protein. J. Dairy Sci. 101:8046-8053.

Lopez, S., J. France, M. S. Dhanoa, F. Mould, and J. Dijkstra. 1999. Comparison of mathematical models to describe disappearance curves obtained using the polyester bag technique for incubating feeds in the rumen. J. Anim. Sci. 77:1875-1888.

Maxin, G., D. R. Ouellet, and H. Lapierre. 2013. Ruminal degradability of dry matter, crude protein, and amino acids in soybean meal, canola meal, com, and wheat dried distillers grains. J. Dairy Sci. 96:5151-5160.

McCleary, B. V., T. S. Gibson, and D. C. Mugford. 1997. Measurement of total starch in cereal products by amyloglucosidase-a-amylase method: Collaborative study. J. AOAC Int. 80:571-579.

McCleary, B. V., M. McNally, and P. Rossiter. 2002. Measurement of resis- tant starch by enzymatic digestion in starch and selected plant materials: Collaborative study. J. AOAC Int. 85:1103-1 111.

Mertens, D. R. 2002. Gravimetric determination of amylase-treated neutral detergent fiber in feeds with refluxing in beakers or crucibles: Collabora- tive study. J. AOAC Int. 85:1217-1240.

Michalet-Doreau, B,, and P. Cemeau. 1991. Influence of foodstuff particle size on in situ degradation of nitrogen in the rumen. Anim. Feed Sci. Technol. 35:69-81.

NorFor. 2011. The Nordic Feed Evaluation System, H. Volden, ed. EAAP Publication No. 130. Wagemngen, The Netherlands: Wageningen Aca- demic Publishers.

NRC (National Research Council). 2001. Nutrient Requirements of Dairy Cattle. 7th rev. ed. Washington, DC: National Academy Press.

Overton, T. R., D. W. LaCount, T. M. Cicela, and J. H. Clark. 1996. Evalu- ation of a ruminally protected methionine product for lactating dairy cows. J. Dairy Sci. 79(4):631-638.

Palmquist, D. L., and T. C. Jenkins. 2003. Challenges with fats and fatty acid methods. J. Anim. Sci. 81:3250-3254.

Paz, H. A., T. J. Klopfenstein, D. Hostetler, S. C. Fernando, E. Castillo- Lopez, and P. J. Kononoff. 2014. Ruminal degradation and intestinal digestibility of protein and amino acids in high-protein feedstuffs com- monly used in dairy diets. J. Dairy Sci. 97:6485-6498.

Ray, P. P, C. Shang, R. O. Maguire, and K. F. Knowlton. 2012. Quantify- ing phytate in dairy digesta and feces: Alkaline extraction ad high- performance ion chromatography. J. Dairy Sci. 95:3248-3258.

Robel, E. J., and A. B. Crane. 1972. An accurate method of correcting unknown amino acid losses from protein hydrolyzates. Anal. Biochem. 48:233-246.

Ross, D. A. 2013. Methods to analyze feeds for nitrogen fractions and digestibility for ruminants with application for the CNCPS. PhD diss., Ithaca, NY: Cornell University.

Rowan, A. M., P. J. Moughan, and M. N. Wilson. 1992. Effect of hydrolysis time on the determination of the amino acid composition of diet, heal digesta, and feces samples and on the determination of dietary amino acid digestibility coefficients. J. Agric. Food Chem. 40(6):981—985.

Rulquin, H., and J. Kowalczyk. 2003. Development of a method for measur- ing lysine and methionine bioavailability in rumen-protected products for cattle. J. Anim. Feed Sci. 12(3):465-474.

Rutherfurd, S. M., and G. S. Gilani. 2009. Amino acid analysis. Curr. Pro- toc. Protein Sci. 58:11.9.1-11.9.37.

Rutherfurd, S. M., P. J. Moughan, D. Lowry, and C. G. Prosser. 2008. Amino acid composition determined using multiple hydrolysis times for three goat milk formulations. Int. J. Food Sci. Nutr. 59:679-690.

Soyeurt, H., F. Dehareng, N. Gengler, S. McParland, E. Wall, D. P. Berry, M. Coffey, and P. Dardenne. 2011. Mid-infrared prediction of bovine milk fatty acids across multiple breeds, production systems, and coun- tries. J. Dairy Sci. 94:1657-1667.

Spears, J. W., K. E. Lloyd, and K. Krafka. 2017. Chromium concentrations in ruminant feed ingredients. J. Dairy Sci. 100:3584-3590.

St-Pierre, N. R., and W. P. Weiss. 2015. Partitioning variation in nutrient composition data of common feeds and mixed diets on commercial dairy farms. J. Dairy Sci. 98:5004-5015.

Sukhija, P. S., and D. L. Palmquist. 1988. Rapid method for determination of total fatty acid content and composition of feedstuffs and feces. J. Agric. Food Chem. 36:1202-1206.

Uden, P. 2006. In vitro studies on microbial efficiency from two cuts of ryegrass (Lolium perenne, cv. Aberdart) with different proportions of sugars and protein. Anim. Feed Sci. Technol. 126:145-156.

Van Soest, P. J. 1965. Use of detergents in analysis of fibrous feeds: III. Study of effects of heating and drying on yield of fiber and lignin in forages. J.AOAC 48:785-790.

Vanzant, E. S., R. C. Cochran, and E. C. Titgemeyer. 1998. Standardization of in situ techniques for ruminant feedstuff evaluation. J. Anim. Sci. 76:2717-2729.

Weinbach, A. P., and D. B. Calvin. 1935. The reducing powers of physi- ologically important carbohydrates. Science 81:407-408.

Weiss, W. P., and N. R. St-Pierre. 2009. A method to quantify changes in supply of metabolizable methionine to dairy cows using concentrations of selenium in milk. J. Dairy Sci. 92:2835-2842.

Weiss, W. P,, C. Hill, and N. St-Pierre. 2014. Proper sampling and sample scheduling can prevent reduced milk yields. Pp. 149-162 in Proceedings of the Tri-State Dairy Nutrition Conference, April 14-16, 2014. https:// 20fd2eal-0814-41a9-868d-fa06f395c8c2.filesusr.com/ugd/36a444_af 115dlc2a3d45069e98491de507261b.pdf. Accessed on July 27, 2021.

Whitehouse, N. L. 2016. Using plasma free amino acid dose response curve method to determine metabolizable protein concentrations of lysine and methionine in rumen protected supplements. PhD diss., University of New Hampshire.

Whitehouse, N. L., C. G. Schwab, and A. F. Brito. 2017. The plasma free amino acid dose-response technique: A proposed methodology for determining lysine relative bioavailability of rumen protected lysine supplements. J. Dairy Sci. 100:9585-9601